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UNIVERSITI TUNKU ABDUL RAHMAN

Faculty of Information and Communication Technology

UCCD1203
DATABASE DEVELOPMENT AND APPLICATIONS

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Marking Section:

Section	Marks
Part A	
Part B	
TOTAL	**/100**

MARKING SCHEME

Case Study: Museum Artifact Management System	
Check list for Part A – Database Design (3.5%)	
1. **Introduction (5 marks)** Describe the business background of the database system including the user of the system and discuss 3 main objectives of this project.	
2. **ER model (15 marks)** You are required to design an ER diagram for the case study given, identify entities, identify relationships, identify associate attribute and determine keys. Check the ERD with the requirements stated in the case. Resolve all the many-many relationships.	
3. **Business Rule Requirements (10 marks)** Identify the business rule involve based on the ERD. Please include any assumptions that you make.	
4. **Report (5 marks)** Submit your report with proper technical report writing skills (cover page, headers & footers, page numbering, headings, sections numbering, proper fonts, spell checked, proper alignments, spacing, marking scheme, table of contents etc).	
SUBTOTAL:	**/35**

Check list for Part B – The Implementation of database using Microsoft Access (6.5%)	
1. **Tables and records (15 marks)** Create all the relations in the ERD and insert the necessary records. (at least 10 records per table except Lookup/composite tables)	
2. **Queries (15 marks)** Create 3 relevant queries based on the given case study and include the corresponding SQL command. Explain the purpose of each query.	
3. **Forms (15 marks)** Create 3 useful forms based on the given case study. You may use subform(s). Explain the purpose of each form. Include colors/images to make your form look interesting.	
4. **Reports (15 marks)** Create 3 useful reports based on the given case study. Explain the purpose of each form. Design your reports so that it will look proper and easy to understand.	
5. **Switchboard (5 marks)** Create a switchboard that includes all the tables, forms, queries and reports. Make sure the switchboard can **start automatically** when the database is open and include the **exit button** to exclude the user from the database.	
SUBTOTAL:	**/65**

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CHAPTER 1 PROJECT BACKGROUND

Project Background

In this chapter, we present the system overview and details about the business background of the Heritage Museum's Museum Artifact Management System. The database system is designed to efficiently manage the collection of artifact information, and the details of targeted users of the database will also be included in this chapter.

The goal of this project is to develop a centralized database system that is user-friendly for the museum administrators to access, store, update, retrieve data and navigate across it.

1.1 System Overview

The Museum Artifact Management System is a centralized database solution designed for the Heritage Museum to streamline the management of artifacts and related operations. The system supports a comprehensive set of functionalities, including storing detailed information about artifacts, tracking their provenance, maintenance history, storage locations, and exhibition records. It also allows users to monitor artifact conditions and manage staff assignments related to conservation, exhibition planning, and curation tasks.

This database system enhances efficiency by enabling museum staff to retrieve and update information quickly, ensure artifact security, and maintain accurate historical records. The system incorporates various entities such as Artifact, Staff, Exhibit, Maintenance, Storage, and Provenance, with clear relationships to facilitate data consistency and integrity. With role-based access and organized data flows, the system aims to support the daily operations of the museum while safeguarding valuable cultural assets.

1.2 Project Objectives

Objective 1:

To develop a structured database system that store artifact information with proper categorization and easy retrieval.

Objective 2:

To allow user to track and monitor the artifacts, including its current location, condition, maintenance record, and exhibition status, to improve artifact data management and security.

Objective 3:

To design an effective resource management sub-system that tracks and optimizes artifact-related resource allocation by integrating staff roles and maintenance workflows, ensuring real-time monitoring of storage facilities

1.3 Target Database User

The targeted users for this database system are archivist, curator, museum conservator, exhibit designer, museum educator, and the human resource department.

An archivist holds a critical role in the museum, as they were required to have exceptional organizational skills as they are responsible for maintaining the data collection of each artifact [1]. Their primary duties allow them to be a user of the database system so that they can catalog each artifact, ensure that the data recorded are organized and have accurate documentation, and provide updates for each record.

A curator holds the highest position among the museum hierarchy, is responsible for acquiring new artifacts. Specifically, this involves conducting research to identify the significant artifacts that are related to the museum's theme, negotiate acquisition with collectors and donors, and seek funding for artifact acquisition [1]. In addition, curators are proficient in using the database system to update and manage the provenance records of each artifact.

A museum conservator is responsible for making preventative conservation measures, such as environmental and climate control, pest management, and retain the aesthetic appeal of artifacts [1]. They need to use the database system to update the condition and maintenance history of each artifact.

An exhibit designer plays a crucial role in planning, design, and executes the layout of museum displays, ensuring that artifacts are presented in a way that is both educational and aesthetically pleasing to enhance the visitors' experience [1]. They need to use the database system to plan and organize future exhibitions features of an artifact.

A museum educator engages with the public and educates visitors as they are responsible for providing guided tours, creating educational materials, and contributing to the museum's educational programs [1]. Educators use the database system to access information on artifacts and exhibits, ensuring they have accurate and up-to-date knowledge to share with visitors.

Lastly, the museum's human resource department also requires access to the database system to record and manage information related to every museum staff member, including their roles and responsibilities.

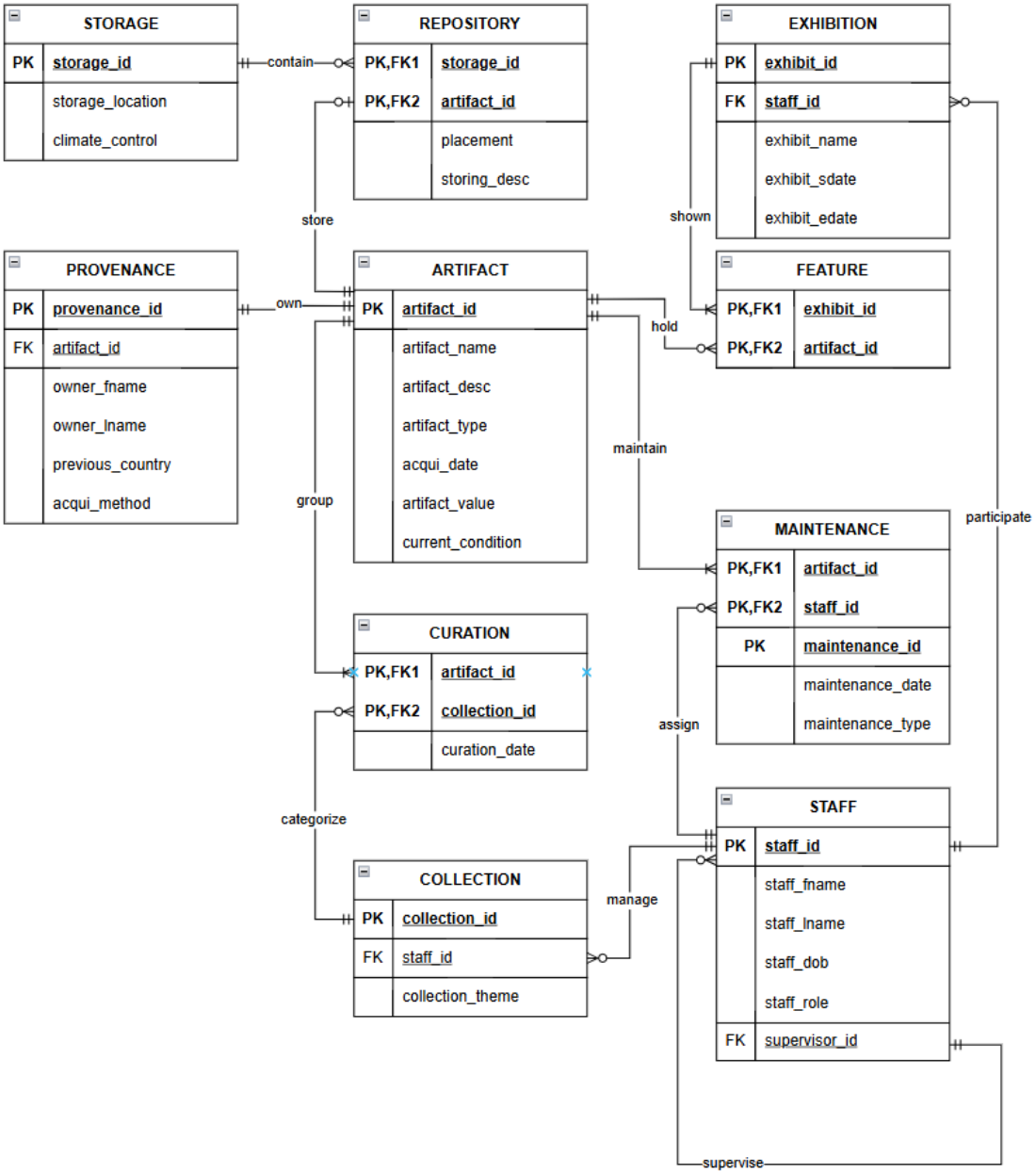
CHAPTER 2 DATABASE DESIGN

2.1 List of Entities and Attributes

Entity	Attributes
ARTIFACT	- **PK** artifact_id - artifact_name - artifact_desc - artifact_type - acqui_date - artifact_value - current_condition
PROVENANCE	- **PK** provenance_id - **FK** artifact_id **reference** ARTIFACT (artifact_id) - owner_fname - owner_lname - previous_country - acqui_method
COLLECTION	- **PK** collection_id - **FK** staff_id **reference** STAFF (staff_id) - collection_theme
CURATION	- **PK, FK** collection_id **reference** COLLECTION (collection_id) - **PK, FK** artifact_id **reference** ARTIFACT (artifact_id) - curation_date
MAINTENANCE	- **PK, FK** staff_id **reference** STAFF (staff_id) - **PK, FK** artifact_id **reference** ARTIFACT (artifact_id) - **PK** maintenance_id - maintenance_date - maintenance_type
STAFF	- **PK** staff_id - staff_fname - staff_lname - staff_dob - staff_role

	- **FK** supervisor_id **reference** STAFF
EXHIBITION	- **PK** exhibit_id - **FK** staff_id **reference** STAFF (staff_id) - exhibit_name - exhibit_sdate - exhibit_edate
STORAGE	- **PK** storage_id - storage_location - climate_control
FEATURE	- **PK, FK** exhibit_id **reference** EXHIBITION - **PK, FK** artifact_id **reference** ARTIFACT
REPOSITORY	- **PK, FK** artifact_id **reference** ARTIFACT - **PK, FK** storage_id **reference** STORAGE - placement - storing_desc

2.2 Entity Relationship Diagram (ERD)



2.3 Assumptions and Business Rules

List of Assumptions

1. Storage

Assume that each storage is uniquely identified.

Assume that a storage can contain multiple repositories.

Assume that climate control and storage capacity are managed per storage unit.

Assume that storage locations do not overlap.

Assume that each artifact is indirectly linked to storage through repositories.

2. Repository

Assume that a repository is stored in exactly one storage.

Assume that each repository stores only one artifact.

Assume that repositories cannot be empty if they exist.

Assume that no repository stores multiple artifacts.

Assume that a repository cannot belong to more than one storage.

3. Artifact

Assume that every artifact may or may not be stored in a repository.

Assume that each artifact has a unique provenance.

Assume that an artifact can have multiple curations and features.

Assume that each artifact may appear in multiple maintenance records.

Assume that artifact location, condition, and related activities are consistently tracked.

4. Provenance

Assume that each provenance is linked to exactly one artifact.

Assume that provenance information is accurate and verified during acquisition.

Assume that provenance data includes acquisition method and previous ownership.

Assume that no artifact can have more than one provenance.

Assume that provenance records are immutable once entered.

5. Curation

Assume that each curation is related to one and only one artifact.

Assume that curations must belong to a valid collection.

Assume that an artifact can have multiple curations over time.

Assume that curation activities are periodically reviewed.

Assume that curation information is necessary for categorizing artifacts.

6. Collection

Assume that a collection manages one or many curations.

Assume that each collection is assigned to exactly one staff member.

Assume that collections do not share artifacts.

Assume that collection themes or categories are clearly defined.

Assume that collections are not auto generated and require staff input.

7. Feature

Assume that a feature is related to exactly one artifact.

Assume that an artifact can have zero or more features.

Assume that features are used to represent artifacts in exhibitions.

Assume that each feature is only linked to one exhibition at a time.

Assume that features provide display and descriptive information about artifacts.

8. Exhibition

Assume that each exhibition displays one or many features.

Assume that an exhibition is managed by a staff member.

Assume that exhibitions are uniquely identified by exhibit_id.

Assume that artifacts can appear in multiple exhibitions through different features.

Assume that exhibitions follow a structured schedule and theme.

9. Staff

Assume that every staff has a unique supervisor except those at the top level (who supervise themselves).

Assume that a staff can be assigned to multiple collections and exhibitions.

Assume that roles include curator, conservator, exhibition manager, and archivist.

Assume that supervisor relationships form a tree hierarchy without cycles.

Assume that a staff may perform maintenance if assigned the conservator role.

10. Maintenance

Assume that maintenance is done by one staff member per record.

Assume that an artifact can have multiple maintenance records over time.

Assume that each maintenance event includes a date and type (e.g., cleaning, repair).

Assume that only staff with the proper role can perform maintenance.

Assume that maintenance records help track artifact condition history.

List of Business Rules (According to ERD)

1. STORAGE AND REPOSIROTY

Each Storage contains one or many repository.

Each Repository contained one and only one Storage.

2. REPOSITORY AND ARTIFACT

Each Repository stores one and only one Artifact.

Each Artifact stored by one or zero Repository.

3. ARTIFACT AND PROVENANCE

Each Artifact owns one and only one Provenance.

Each Provenance owned by one and only one Artifact.

4. ARTIFACT AND CURATION AND COLLECTION

4.1 ARTIFACT AND CURATION

Each Artifact group one or many Curation.

Each Curation grouped by one and only one Artifact.

4.2 CURATION AND COLLECTION

Each Curation categorized by one and only one Collection.

Each Collection categorize to zero or many Curation.

5. ARTIFACT AND FEATURE AND EXHIBITION

5.1 ARTIFACT AND FEATURE

Each Artifact is held by zero or many Features.

Each Feature hold one and only one Artifact.

5.2 FEATURE AND EXHIBITION

Each Feature shown by one and only one Exhibition.

Each Exhibition shows one or many Features .

6. COLLECTION AND STAFF

Each Collection managed by one and only one Staff.

Each Staff manage zero or many Collection .

7. EXHIBITION AND STAFF

Each Exhibition participated by one and only one Staff.

Each Staff participates in zero or many Exhibition.

8. STAFF

Each Staff supervised by one and only one Staff which is a Supervisor.

Each Supervisor that is a staff and supervises zero or many Staff.

9. STAFF AND MAINTENANCE

Each Staff get assigned to zero or many Maintenance

Each Maintenance assigned by one and only one staff

10. MAINTENANCE AND ARTIFACT

Each Maintenance will maintain one and only one Artifact

Each Artifact will be maintained by one or many times of maintenance.

CHAPTER 3 DATABASE DEVELOPMENT

3.1 List of Tables

In this project, the patient, guardian, doctor, duty slot, diagnosis, treatment, medication, and treatment receipt are among the tables that were used. There are a total of 8 tables in the project, each of which contains a collection of named columns and an unspecified number of rows to store data and information. Each column has a set data type, and the values that can be entered into each field are restricted by domain constraints such as check constraints, not null constraints, unique constraints, and default constraints.

3.1.1 Data Dictionary and Table Record

a) ARTIFACT

- Data Dictionary

ARTIFACT

Field Name	Data Type	Description (Optional)
artifact_id	AutoNumber	Unique identifier for each artifact (Primary key)
artifact_name	Short Text	The official name or title of the artifact.
artifact_desc	Long Text	A detailed textual description of the artifact, including its origin, cultural significance, creation period, and historical background.
artifact_type	Short Text	The category or classification of the artifact, such as Painting, Sculpture, or Manuscript, indicating its artistic or historical medium/form.
acqui_date	Date/Time	The date the artifact was acquired by the museum. Useful for tracking ownership history and cataloging.
artifact_value	Currency	The estimated or appraised monetary value of the artifact. This could be used for insurance, inventory, or financial reporting.
current_condition	Short Text	The current physical condition or state of the artifact.

Figure 3.1.1.1: Data dictionary of ARTIFACT entity

- Table Records

ARTIFACT

artifact_id	artifact_name	artifact_desc	artifact_type	acqui_date	artifact_value	current_cond
a001	Mona Lisa	The Mona Lisa is a world-renowned port	Painting	2005-03-15	$8,000,000	Excellent
a002	Rosetta Stone	The Rosetta Stone is a granodiorite stele	Manuscript	1998-07-20	$2,000,000	Good
a003	Venusde Milo	The Venus de Milo is an ancient Greek s	Sculpture	2001-11-01	$5,000,000	Good
a004	Sun Tzu Manuscript	The Sun Tzu Manuscript is an ancient Cl	Manuscript	2012-09-10	$3,000,000	Good
a005	The Starry Night	The Starry Night is a famous oil painting	Painting	2007-06-21	$7,500,000	Excellent
a006	Dead Sea Scroll	The Dead Sea Scrolls are a collection of	Manuscript	2015-12-01	$4,000,000	Good
a007	Terracotta Warrior	The Terracotta Warriors are life-sized cla	Sculpture	2003-04-18	$3,500,000	Need Repair
a008	Liberty Bell Replica	The Liberty Bell Replica is a faithful repr	Sculpture	2000-08-10	$1,000,000	Good
a009	Egyptian Sarcophagus	An Egyptian sarcophagus is a coffin typi	Sculpture	1995-10-25	$4,500,000	Excellent
a010	Gutenberg Bible Page	A page from the Gutenberg Bible, printe	Manuscript	2018-01-30	$1,500,000	Need Repair
a011	The Scream	The Scream is a famous painting by Nor	Painting	2006-11-11	$8,200,000	Excellent
a012	Roman Coin Set	A Roman coin set consists of a collectior	Sculpture	2010-02-14	$1,200,000	Good
a013	Aztec Calendar Stone	The Aztec Calendar Stone, also known a	Sculpture	2008-07-07	$6,000,000	Good
a014	Baylex Tapestry	The Bayeux Tapestry is an embroidered	Manuscript	2013-03-28	$2,200,000	Good
a015	The Great Wave Print	The Great Wave off Kanagawa is a wood	Painting	2011-05-19	$1,800,000	Need Repair

Figure 3.1.1.2: Table Records of ARTIFACT entity

b) COLLECTION

- Data Dictionary

COLLECTION

Field Name	Data Type	Description (Optional)
collection_id	AutoNumber	A unique identifier for each collection. (Primary Key)
staff_id	Number	References the ID of the staff member with role curator. (Foreign Key references STAFF)
collection_theme	Short Text	Describes the main theme or focus of the collection.

Figure 3.1.2.1: Data dictionary of COLLECTION entity

- Table Records

COLLECTION

collection_id	staff_id	collection_theme
c001	s003	Ancient Civilizations
c002	s003	Renaissance Art
c003	s015	Masterpieces
c004	s013	Cultural Heritage

Figure 3.1.2.2: Table Records of COLLECTION entity

c) CURATION

- Data Dictionary

CURATION

Field Name	Data Type	Description (Optional)
artifact_id	Number	References the ID of the artifact being curated. (Foreign Key references ARTIFACT)
collection_id	Number	References the ID of the collection where the artifact is being curated. (Foreign Key references COLLECTION)
curation_date	Date/Time	The date when the artifact was added to or curated within the collection.

Figure 3.1.3.1: Data dictionary of CURATION entity

- Table Records

CURATION

artifact_id	collection_id	curation_date
a001	c002	2005-04-01
a001	c003	2005-04-03
a002	c001	1999-01-15
a002	c004	1999-01-18
a003	c001	2002-01-10
a003	c003	2002-01-12
a004	c002	2012-10-01
a005	c003	2007-07-01
a006	c001	2016-01-05
a006	c004	2016-01-07
a007	c001	2003-05-01
a008	c004	2000-09-05
a009	c001	1996-01-10
a010	c002	2018-02-15
a010	c004	2018-02-17
a011	c003	2006-12-01
a012	c001	2010-03-01
a013	c001	2008-08-01
a014	c004	2013-04-01
a015	c003	2011-06-01

Figure 3.1.3.2: Table Records of CURATION entity

d) EXHIBITION

- Data Dictionary

EXHIBITION

Field Name	Data Type	Description (Optional)
exhibit_id	AutoNumber	Unique identifier for each exhibition. (Primary key)
staff_id	Number	References the staff responsible for the exhibition. (Foreign Key references STAFF)
exhibit_name	Short Text	The name of the exhibition.
exhibit_sdate	Date/Time	The start date of the exhibition.
exhibit_edate	Date/Time	The end date of the exhibition.

Figure 3.1.4.1: Data dictionary of EXHIBITION entity

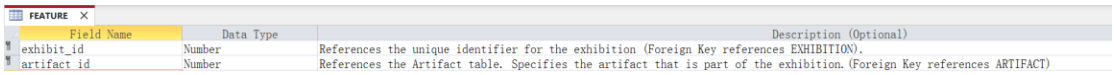
EXHIBITION

exhibit_id	staff_id	exhibit_name	exhibit_sdate	exhibit_edate
e001	s007	Masterpieces of the Mode	2005-03-15	2007-06-21
e002	s008	Civilizations of the Past	1998-07-20	1998-07-20
e003	s008	Ancient Wonders	2007-06-21	2012-09-10
e004	s007	Renaissance Masterpieces	2005-03-15	2012-09-10
e005	s012	Global Heritage	2015-12-01	2018-01-30

Figure 3.1.4.2: Table Records of EXHIBITION entity

e) FEATURE

- Data Dictionary



FEATURE

Field Name	Data Type	Description (Optional)
exhibit_id	Number	References the unique identifier for the exhibition (Foreign Key references EXHIBITION).
artifact_id	Number	References the Artifact table. Specifies the artifact that is part of the exhibition. (Foreign Key references ARTIFACT)

Figure 3.1.5.1: Data dictionary of FEATURE entity

- Table Records

FEATURE

exhibit_id	artifact_id
e001	a001
e001	a002
e002	a002
e003	a001
e003	a003
e003	a005
e003	a010
e003	a013
e004	a001
e004	a002
e004	a003
e004	a004
e004	a011
e004	a015
e005	a004
e005	a005
e005	a006
e005	a011
e005	a012

Figure 3.1.5.2: Table Records of FEATURE entity

f) MAINTENANCE

- Data Dictionary

MAINTENANCE

Field Name	Data Type	Description (Optional)
artifact_id	Number	References the Artifact table. Specifies the artifact undergoing maintenance. (Foreign Key references ARTIFACT)
maintenance_id	Short Text	Unique identifier for each maintenance record.
staff_id	Number	References the Staff table. Identifies the staff member responsible for the maintenance. (Foreign Key references STAFF)
maintenance_date	Date/Time	The date when the maintenance was performed.
maintenence_type	Short Text	Type of maintenance performed

Figure 3.1.6.1: Data dictionary of MAINTENANCE entity

MAINTENANCE

artifact_id	maintenance_id	staff_id	maintenance_date	maintenence_type
a001	1	s004	2006-04-12	Cleaning
a001	2	s004	2022-01-13	Repair
a001	3	s004	2007-05-30	Repair
a006	1	s005	2017-10-21	Cleaning
a006	2	s005	2018-09-14	Restoration
a007	1	s005	2004-06-18	Inspection
a009	1	s006	1996-05-23	Cleaning
a012	1	s011	2012-07-11	Preservation

Figure 3.1.6.2: Table Records of MAINTENANCE entity

g) PROVENANCE

- Data Dictionary

PROVENANCE

Field Name	Data Type	Description (Optional)
provenance_id	Short Text	Unique identifier for each provenance record.
artifact_id	Number	References the Artifact table. Specifies which artifact this provenance record belongs to. (Foreign Key references ARTIFACT)
owner_fname	Short Text	First name of the artifact' s previous owner.
owner_lname	Short Text	Last name of the artifact' s previous owner.
previous_country	Short Text	Country where the artifact was previously held.
acqui_method	Short Text	Method of acquisition, such as Donation, Purchase, or Excavation.

Figure 3.1.7.1: Data dictionary of PROVENANCE entity

- Table Records

PROVENANCE

provenance_id	artifact_id	owner_fname	owner_lname	previous_country	acqui_method
p001	a001	Carlo	Gatti	Italy	Purchase
p002	a002	Isabella	Moreno	Spain	Donation
p003	a003	Henry	Cavendish	England	Purchase
p004	a004	Amara	Hassan	Egypt	Excavation
p005	a005	Klaus	Muller	Germany	Donation
p006	a006	Elise	Dubois	France	Purchase
p007	a007	Hiroshi	Takeda	Japan	Purchase
p008	a008	Raj	Mehta	India	Donation
p009	a009	Sarah	Thompson	USA	Inheritance
p010	a010	Mei	Zhang	China	Excavation
p011	a011	Laila	Saeed	Iraq	Donation
p012	a012	Giovanni	Rossi	Italy	Purchase
p013	a013	Olivia	Bennett	Australia	Donation
p014	a014	Ahmed	Farouk	Egypt	Excavation
p015	a015	Frederic	Lefevre	France	Purchase

Figure 3.1.7.2: Table Records of PROVENANCE entity

h) REPOSITORY

- Data Dictionary

REPOSITORY

Field Name	Data Type	Description (Optional)
storage_id	Number	Unique identifier for each storage record.
artifact_id	Number	References the Artifact table. Specifies the artifact being stored. (Foreign Key references ARTIFACT)
placement	Short Text	The storage placement for the artifact (e.g., Display Case, Cabinet, Pedestal, Rack).
storing_desc	Long Text	Description of how the artifact is stored, including conditions such as climate control, lighting, and material type.

Figure 3.1.8.1: Data dictionary of REPOSITORY entity

- Table Records

REPOSITORY

storage_id	artifact_id	placement	storing_desc
s001	a001	Display Case	Store in a glass display case with UV protection and climate control.
s001	a002	Cabinet	Keep in a wooden cabinet with climate control and low traffic.
s001	a004	Cabinet	Store in a climate-controlled cabinet with acid-free materials.
s001	a005	Pedestal	Display on a pedestal with controlled lighting.
s001	a006	Cabinet	Store in a glass-fronted cabinet with climate control.
s001	a010	Cabinet	Store in a locked, climate-controlled cabinet with acid-free materials.
s001	a011	Pedestal	Display on a pedestal with controlled lighting and stable climate.
s001	a014	Cabinet	Store flat in a climate-controlled, dark cabinet.
s001	a015	Pedestal	Display on a pedestal with controlled lighting and stable environment.
s003	a007	Rack	Store on a rack in a stable temperature and humidity environment.
s004	a003	Rack	Place on a sturdy rack in a vibration-free area.
s004	a008	Display Case	Keep in a display case with controlled climate and minimal handling.
s004	a009	Display Case	Place in a sealed case with climate control and away from sunlight.
s004	a012	Cabinet	Store in a cabinet with compartments and climate control.
s004	a013	Rack	Place on a sturdy rack in a vibration-free, temperature-stable area.

Figure 3.1.8.2: Table Records of REPOSITORY entity

i) STAFF

- Data Dictionary

STAFF

Field Name	Data Type	Description (Optional)
staff_id	AutoNumber	Unique identifier for each staff member.
staff_fname	Short Text	First name of the staff member.
staff_lname	Short Text	Last name of the staff member.
staff_dob	Date/Time	Date of birth of the staff member.
staff_role	Short Text	The role of the staff member
supervisor_id	Number	References the staff member' s supervisor. (Foreign Key references STAFF)

Figure 3.1.9.1: Data dictionary of STAFF entity

- Table Records

STAFF

staff_id	staff_fname	staff_lname	staff_dob	staff_role	supervisor_id
s001	John	Smith	1980-05-15	Curator	s001
s002	Jane	Doe	1975-09-30	Curator	s001
s003	Robert	Brown	1985-02-20	Curator	s001
s004	Alice	Green	1990-03-10	Conservator	s002
s005	Micheal	White	1982-07-25	Conservator	s004
s006	Emily	Black	1988-11-12	Conservator	s011
s007	David	Williams	1978-01-05	Exhibition Manager	s013
s008	Sarah	Johnson	1993-06-18	Exhibition Manager	s013
s009	William	Harris	1983-08-30	Archivist	s014
s010	Jessica	Clark	1991-04-03	Archivist	s014
s011	Olivia	Martinez	1986-10-11	Conservator	s002
s012	Daniel	Moore	1994-01-22	Exhibition Manager	s013
s013	Thomas	Lee	1980-05-25	Curator	s003
s014	Sophia	King	1992-12-02	Archivist	s015
s015	Charles	Scott	1987-02-15	Curator	s003

Figure 3.1.9.2: Table Records of STAFF entity

j) STORAGE

- Data Dictionary

STORAGE

Field Name	Data Type	Description (Optional)
storage_id	AutoNumber	Unique identifier for each storage location.
storage_location	Short Text	Name of the storage location
climate_control	Short Text	Specifies the climate control features available in the storage location

Figure 3.1.10.1: Data dictionary of STORAGE entity

- Table Records

STORAGE

storage_id	storage_location	climate_control
s001	North Room	Humidity & Temperature Control
s002	South Room	Temperature Control
s003	East Room	Cold & Dry
s004	West Room	Air Purified

Figure 3.1.10.2: Table Records of STORAGE entity

3.2 Switchboard

The Switchboard in Microsoft Access is a user-friendly interface that serves as a central navigation hub for users to access various forms, reports, and queries within the database system. It simplifies navigation by organizing and linking key functions into a single, easy-to-use interface, allowing users to quickly perform tasks without needing to navigate through multiple menus or objects. This feature enhances efficiency, especially for non-technical users, by providing a clear and straightforward way to interact with the database. In this project, we use a switchboard to case the navigation process for the staff members in The Heritage Museum houses.

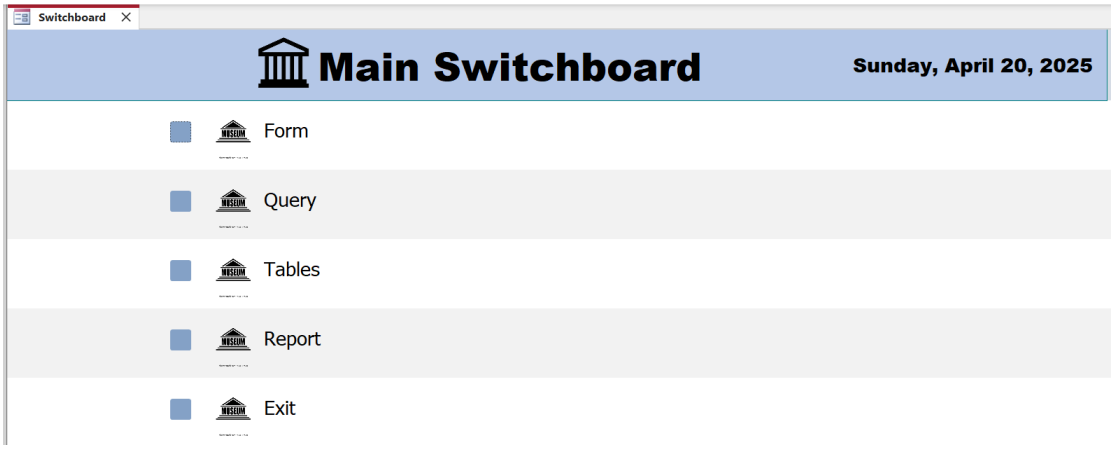


Figure 3.2.1: Main Switchboard

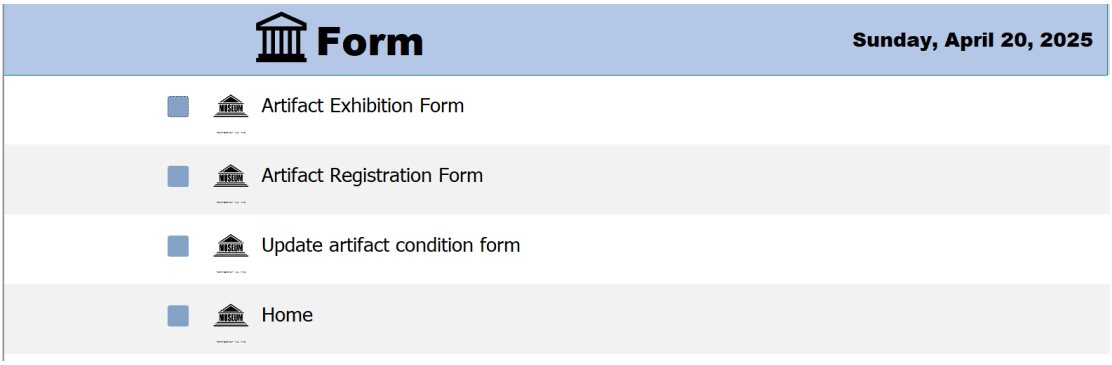


Figure 3.2.2: Form Switchboard

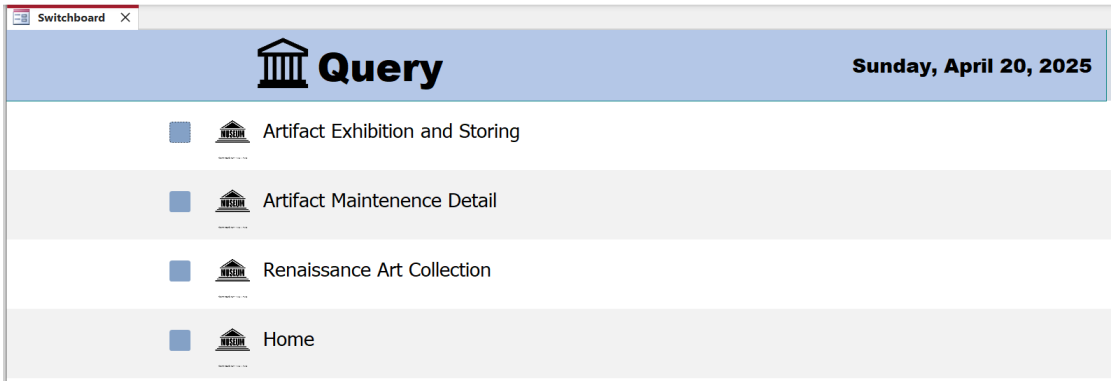


Figure 3.2.3: Query Switchboard

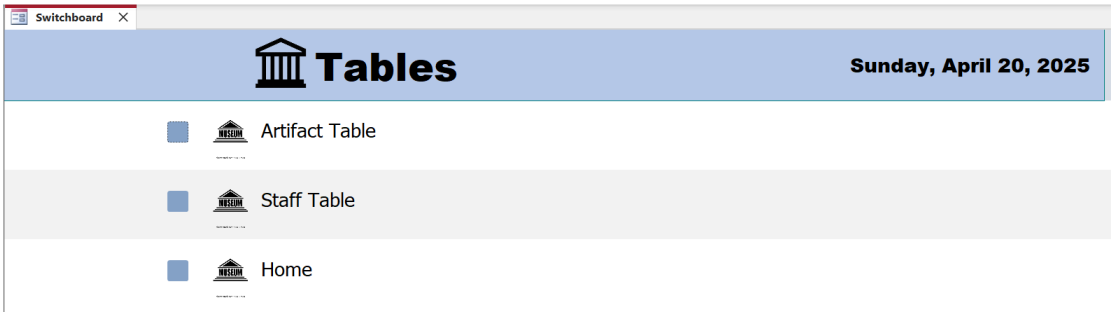


Figure 3.2.4: Table Switchboard

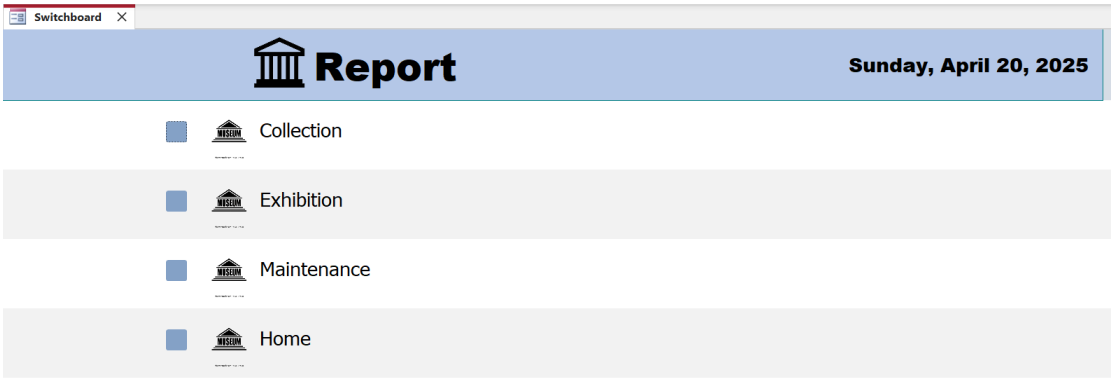


Figure 3.2.5: Report Switchboard

CHAPTER 4 DATABASE OBJECTS

Database Objects

A database object is a structured component within a database that serves a specific function. Examples of database objects include tables, queries, forms, reports, macros, and modules. Each object plays a distinct role and contributes to the overall design, functionality, and integrity of the database system.

4.1 Queries

Queries are used to extract, filter, and analyze data from one or more tables within a database. They allow users to retrieve specific information based on defined criteria, making it easier to focus on relevant data. By streamlining data retrieval, queries help users save time and enhance productivity in database management.

4.1.1 Query 1

Explanation:

The "*Renaissance Art Collection"* query is created to search for all artifacts that belong to the 'Renaissance Art' collection. Information such as the artifact's ID, name, and type will be displayed if the artifact is part of the specified collection. Since an artifact may belong to more than one collection, the query filters to show only those related to 'Renaissance Art'. It serves to help museum staff efficiently access and organize specific artifacts for exhibition planning, inventory review, or thematic research purposes.

SQL command:

```
SELECT COLLECTION.collection_id, COLLECTION.collection_theme,
ARTIFACT.artifact_id, ARTIFACT.artifact_name, ARTIFACT.artifact_type,
CURATION.curation_date
FROM (COLLECTION INNER JOIN CURATION ON COLLECTION.collection_id
= CURATION.collection_id) INNER JOIN ARTIFACT ON CURATION.artifact_id
= ARTIFACT.artifact_id
WHERE COLLECTION.collection_theme = "Renaissance Art";
```

Sample query 1:

Renaissance Art Collection

collection_id	collection_theme	artifact_id	artifact_name	artifact_type	curation_date
c002	Renaissance Art	a001	Mona Lisa	Painting	2005-04-01
c002	Renaissance Art	a004	Sun Tzu Manuscript	Manuscript	2012-10-01
c002	Renaissance Art	a010	Gutenberg Bible Page	Manuscript	2018-02-15

Figure 4.1.1.1: "Renaissance Art Collection" query

4.1.2 Query 2

Explanation:

The "*Artifact Maintenance Detail"* query is created to retrieve all artifacts along with their maintenance records and the name of the staff member responsible for each task. The information displayed includes the artifact's details, maintenance type, maintenance date, and staff name. An artifact may undergo multiple maintenance activities over time, so all related records are shown together. This query enables the museum staff to track artifact care history and assign responsibility, promoting preservation efforts and ensuring long-term artifact integrity.

SQL command:

```
SELECT ARTIFACT.artifact_id, ARTIFACT.artifact_name,
ARTIFACT.artifact_type, MAINTENANCE.maintenance_id,
MAINTENANCE.maintenance_date, MAINTENANCE.maintenance_type,
STAFF.staff_id, STAFF.staff_fname, STAFF.staff_lname
FROM (ARTIFACT INNER JOIN MAINTENANCE ON ARTIFACT.artifact_id =
MAINTENANCE.artifact_id) INNER JOIN STAFF ON MAINTENANCE.staff_id =
STAFF.staff_id;
```

Sample query 2:

Artifact Maintenence Detail

artifact_id	artifact_name	artifact_type	maintenance_id	maintenance_date	maintenence_type	staff_id	staff_fname	staff_lname
a001	Mona Lisa	Painting	1	2006-04-12	Cleaning	s004	Alice	Green
a001	Mona Lisa	Painting	2	2022-01-13	Repair	s004	Alice	Green
a001	Mona Lisa	Painting	3	2007-05-30	Repair	s004	Alice	Green
a006	Dead Sea Scroll	Manuscript	1	2017-10-21	Cleaning	s005	Micheal	White
a006	Dead Sea Scroll	Manuscript	2	2018-09-14	Restoration	s005	Micheal	White
a007	Terracotta Warrior	Sculpture	1	2004-06-18	Inspection	s005	Micheal	White
a009	Egyptian Sarcophagus	Sculpture	1	1996-05-23	Cleaning	s006	Emily	Black
a012	Roman Coin Set	Sculpture	1	2012-07-11	Preservation	s011	Olivia	Martinez

Figure 4.1.2.1: "Artifact Maintenance Detail" query

4.1.3 Query 3

Explanation:

The "*Artifact Exhibition and Storing*" query is created to display all artifacts together with their exhibition and storage details. The information includes artifact names, associated exhibitions (if any), and storage locations. Some artifacts may not currently be part of any exhibition, and in such cases, the exhibition fields will display a dash instead of being left blank. This query helps museum staff to gain a comprehensive overview of the location and display status of each artifact, supporting space planning and effective collection management.

SQL command:

```
SELECT ARTIFACT.artifact_id, ARTIFACT.artifact_name,
ARTIFACT.artifact_type, EXHIBIT.exhibit_id, EXHIBIT.exhibit_name,
STORAGE.storage_id, STORAGE.storage_location
FROM ARTIFACT LEFT JOIN EXHIBIT_ARTIFACT ON ARTIFACT.artifact_id =
EXHIBIT_ARTIFACT.artifact_id
LEFT JOIN EXHIBIT ON EXHIBIT_ARTIFACT.exhibit_id = EXHIBIT.exhibit_id
LEFT JOIN STORAGE ON ARTIFACT.artifact_id = STORAGE.artifact_id;
```

Sample query 3:

Artifact Exhibition and Storing

artifact_id	artifact_name	artifact_type	exhibit_id	exhibit_name	storage_id	storage_location
a001	Mona Lisa	Painting	e001	Masterpieces of the Modern Era	s001	North Room
a001	Mona Lisa	Painting	e003	Ancient Wonders	s001	North Room
a001	Mona Lisa	Painting	e004	Renaissance Masterpieces	s001	North Room
a002	Rosetta Stone	Manuscript	e001	Masterpieces of the Modern Era	s001	North Room
a002	Rosetta Stone	Manuscript	e002	Civilizations of the Past	s001	North Room
a002	Rosetta Stone	Manuscript	e004	Renaissance Masterpieces	s001	North Room
a004	Sun Tzu Manuscript	Manuscript	e004	Renaissance Masterpieces	s001	North Room
a004	Sun Tzu Manuscript	Manuscript	e005	Global Heritage	s001	North Room
a005	The Starry Night	Painting	e003	Ancient Wonders	s001	North Room
a005	The Starry Night	Painting	e005	Global Heritage	s001	North Room
a006	Dead Sea Scroll	Manuscript	e005	Global Heritage	s001	North Room
a010	Gutenberg Bible Page	Manuscript	e003	Ancient Wonders	s001	North Room
a011	The Scream	Painting	e004	Renaissance Masterpieces	s001	North Room
a011	The Scream	Painting	e005	Global Heritage	s001	North Room
a014	Baylex Tapestry	Manuscript			s001	North Room
a015	The Great Wave Print	Painting	e004	Renaissance Masterpieces	s001	North Room
a007	Terracotta Warrior	Sculpture			s003	East Room
a003	Venusde Milo	Sculpture	e003	Ancient Wonders	s004	West Room
a003	Venusde Milo	Sculpture	e004	Renaissance Masterpieces	s004	West Room
a008	Liberty Bell Replica	Sculpture			s004	West Room
a009	Egyptian Sarcophagus	Sculpture			s004	West Room
a012	Roman Coin Set	Sculpture	e005	Global Heritage	s004	West Room
a013	Aztec Calendar Stone	Sculpture	e003	Ancient Wonders	s004	West Room

Figure 4.1.3.1: "Artifact Exhibition and Storing" query

4.2 Forms

A form is a customized window or screen that allows users to interact with the data in an Access database without directly opening the tables. It can display one or multiple records simultaneously and often includes text boxes, combo boxes, buttons, and other controls.

4.2.1 Form 1

The **Artifact Registration Form** contains all the detailed information about each artefact, including the artifact ID, artifact name, artifact description, acquisition date, and artifact value. Users can navigate through the records using the **Previous** and **Next** buttons, as shown in *Figure 4.2.1.1*. Additionally, users can also edit and delete the existing records using the respective **Delete** and **Register** buttons. Besides, users can also add new artifacts by filling out the form as shown in *Figure 4.2.1.2* and clicking the **Register** button to save the new entry. The data will automatically update to the artifact table after clicking the **Register** button.

Sample Form 1:

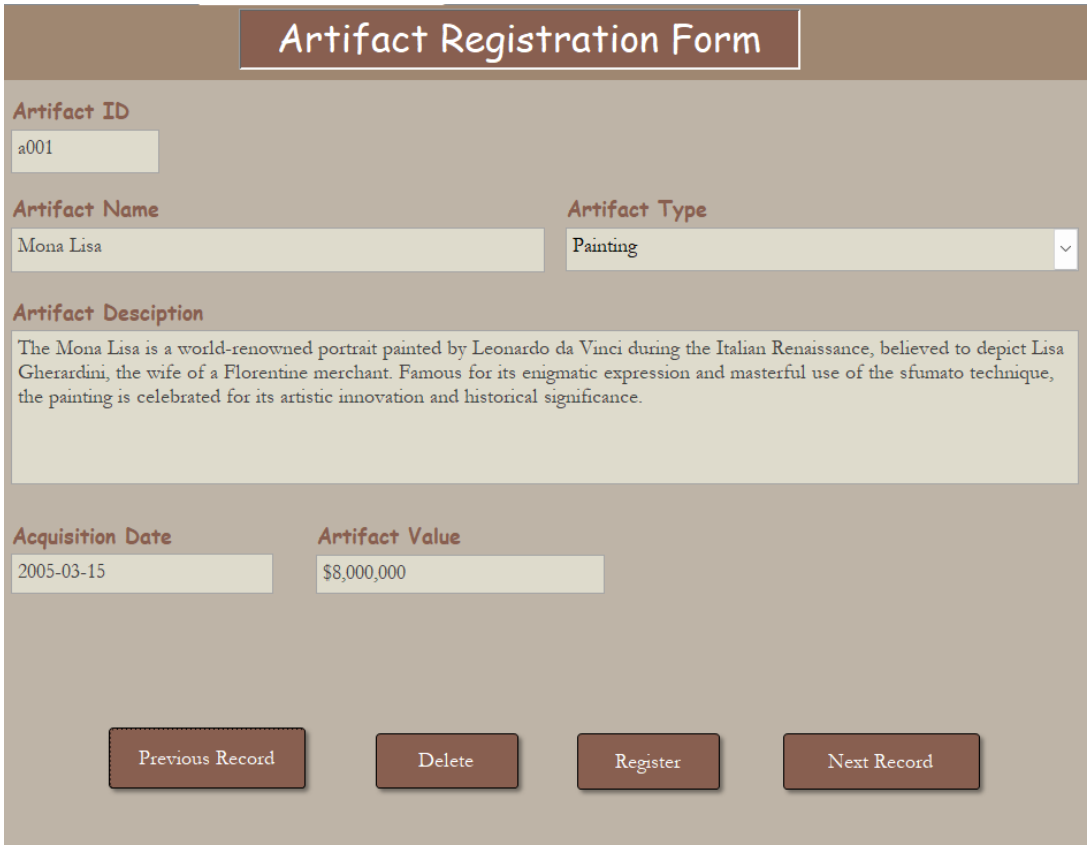


Figure 4.2.1.1 Artifact Record Navigation and Management Interface

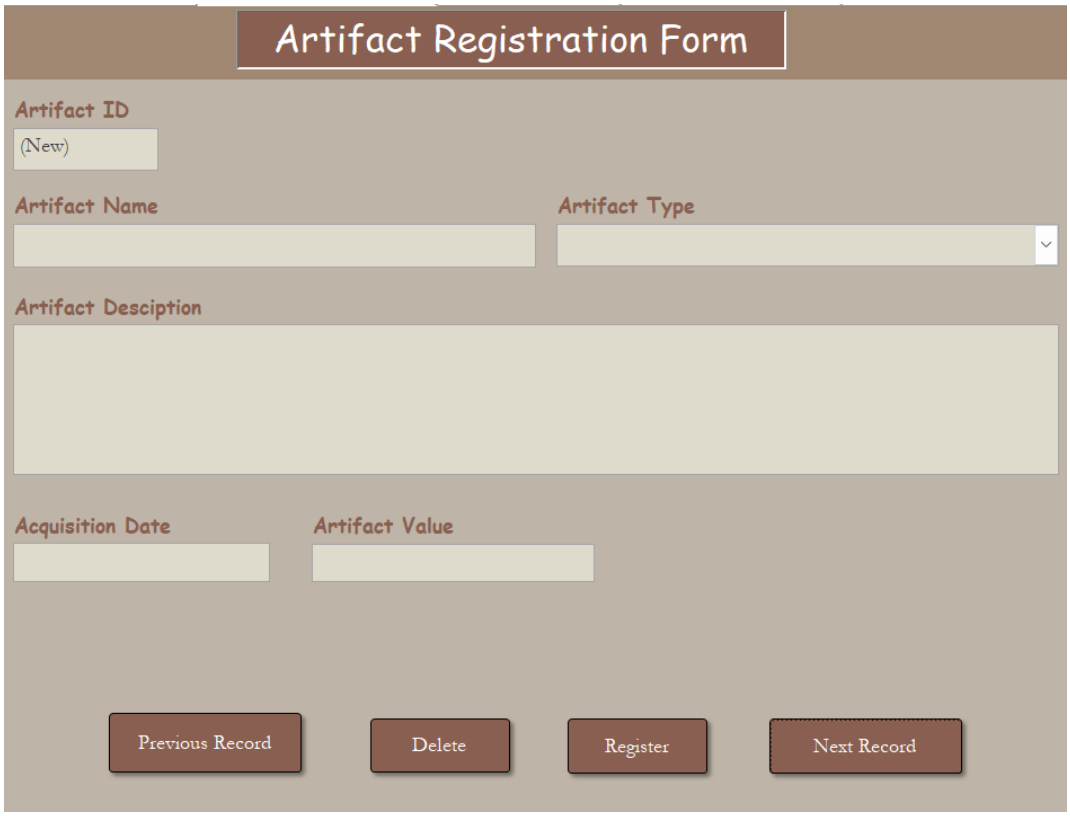


Figure 4.2.1.3 New Artifact Registration Form

4.2.2 Form 2

The Update Artifact Condition Form allows users to view the maintenance history and update the current condition for each artifact. The artifact ID and artifact name fields are locked and cannot be edited. Users are only allowed to change the **current condition** by selecting one of the three predefined options from the combo box, as shown in *Figure 4.2.2.1*.

Sample Form 2:

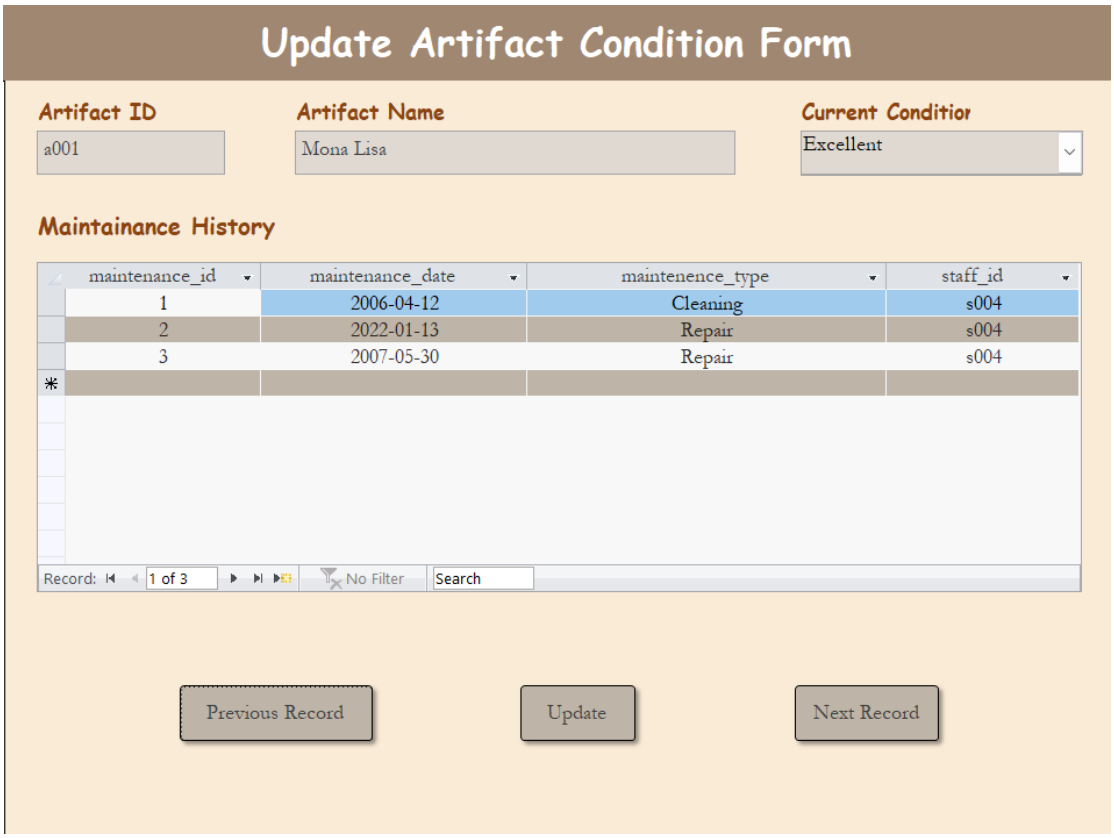


Figure 4.2.2.1 Artifact Condition Update and Maintenance History Form

4.2.3 Form 3

The Artifact Exhibition Form shows details information of each exhibition, including exhibition ID, exhibition name, start and end date, the artifact involved, and the staff involved. Users can view all exhibition details and create new records, as shown in *Figure 4.2.3.1*.

Sample Form 3:

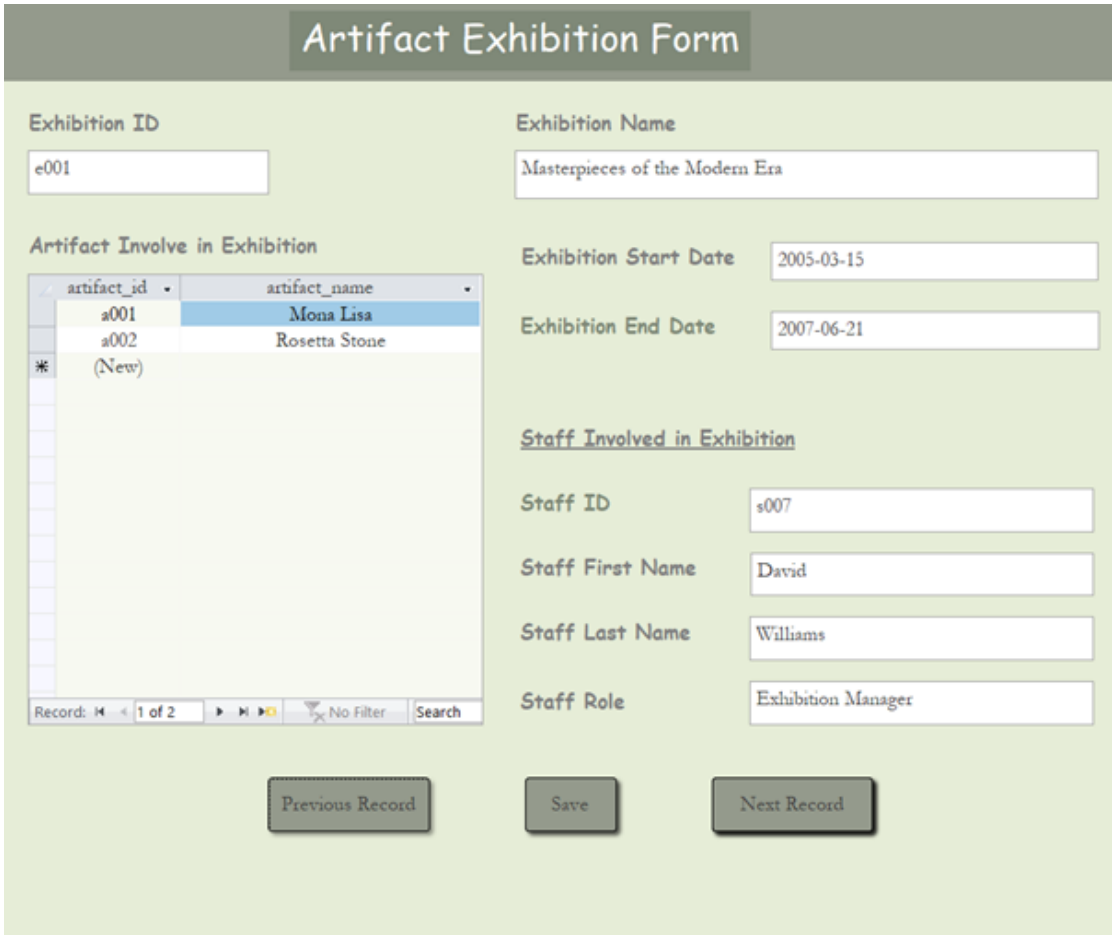


Figure 4.2.3.1 Exhibition Details and Management Form

4.3 Report

Reports are used for presenting summarized and structured information stored in a database. For The Heritage Museum, reports enable staff members to extract essential details about artifacts, their origins, and maintenance history in a clear, concise, and meaningful format. These reports are especially valuable for museum curators and management, as they provide a comprehensive overview of the collection, aiding in preservation planning, historical research, and decision-making regarding exhibitions. Reports can be generated for specific categories of artifacts or for the entire collection and can be customized to present data in a format that is easy to interpret and use.

4.3.1 Report 1

Explanation:

The Artifact Collection Report below demonstrates the organization and status of the museum's holdings by grouping each item under its respective collection theme. By categorizing artifacts such as the Egyptian Sarcophagus and the Gutenberg Bible Page under themes like Ancient Civilizations and Renaissance Art, the report makes it easy to track acquisition dates and monitor the growth of each collection over time. It presents essential fields like Collection Theme, Artifact Name, and Acquisition Date in a clear, tabular layout, allowing curators and managers to quickly assess when key pieces entered the collection and to identify gaps or recent additions. This report is a vital tool for the museum's leadership, supporting informed decision about preservation scheduling, exhibition planning, and future acquisitions. Reports can be generated for individual themes or for the entire collection and customized to highlight specific date ranges or artifacts categories in a format that is straightforward to interpret and act upon.

Sample Report 1:

Collection Report Sunday, April 20, 2025

Collection Theme	Artifact Name	Acqui Date
Ancient Civilizations		
	Aztec Calendar Stone	2008-07-07
	Dead Sea Scroll	2015-12-01
	Egyptian Sarcophagus	1995-10-25
	Roman Coin Set	2010-02-14
	Rosetta Stone	1998-07-20
	Terracotta Warrior	2003-04-18
	Venusde Milo	2001-11-01
Renaissance Art		
	Gutenberg Bible Page	2018-01-30
	Mona Lisa	2005-03-15
	Sun Tzu Manuscript	2012-09-10
Masterpieces		
	Mona Lisa	2005-03-15
	The Great Wave Print	2011-05-19
	The Scream	2006-11-11
	The Starry Night	2007-06-21
	Venusde Milo	2001-11-01
Cultural Heritage		
	Baylex Tapestry	2013-03-28
	Dead Sea Scroll	2015-12-01
	Gutenberg Bible Page	2018-01-30
	Liberty Bell Replica	2000-08-10
	Rosetta Stone	1998-07-20

Sunday, April 20, 2025 Page 1 of 1

Figure 4.3.1.1: Collection Report

4.3.2 Report 2

Explanation:

The Exhibition Report provides a comprehensive overview of museum or gallery exhibitions by tracking essential details such as exhibit names, dates, and associated artifacts. Structured to manage the logistics of artifact display, this report ensures efficient scheduling by documenting the start and end dates of each exhibit alongside the specific artifacts included. In Microsoft Access, this report is crucial for preventing conflicts where the same artifact might be double-booked across overlapping exhibits, while also aiding in historical analysis and future planning. By centralizing data on artifact utilization and exhibit timelines, administrators can optimize resource allocation, assess artifact popularity, and streamline exhibit rotations. This structured approach supports informed decision-making for curators and managers, enhancing operational efficiency and ensuring seamless coordination of exhibitions.

Sample Report 2



Exhibitation Report Sunday, April 20, 2025

Exhibit Name	Start Date	End Date	Artifact Name
Ancient Wonders	2007-06-21	2012-09-10	
			Mona Lisa
			Venusde Milo
			Aztec Calendar Stone
			Gutenberg Bible Page
			The Starry Night
Civilizations of the Past	1998-07-20	1998-07-20	
			Rosetta Stone
Global Heritage	2015-12-01	2018-01-30	
			The Scream
			Dead Sea Scroll
			The Starry Night
			Sun Tzu Manuscript
			Roman Coin Set
Masterpieces of the Modern Era	2005-03-15	2007-06-21	
			Rosetta Stone
			Mona Lisa
Renaissance Masterpieces	2005-03-15	2012-09-10	
			Venusde Milo
			Rosetta Stone
			Sun Tzu Manuscript
			The Scream
			The Great Wave Print
			Mona Lisa

Sunday, April 20, 2025 Page 1 of 1

Figure 4.3.2.1: Exhibition Report

4.3.3 Report 3

Explanation:

The Maintenance Report systematically tracks the preservation activities and condition status of museum artifacts, categorizing them by type (e.g., Manuscript, Painting, Sculpture) and documenting critical details such as maintenance type, artifact condition, and maintenance dates. For example, it records repeated repairs and cleanings for the *Mona Lisa* and highlights artifacts requiring urgent attention, like the *Terracotta Warrior* marked as "Need Repair." In Microsoft Access, this report is vital for ensuring artifact longevity by scheduling timely interventions, avoiding maintenance overlaps with exhibitions and maintaining historical longevity efforts. It aids in resource allocation, budgeting for repairs, and preventing deterioration by flagging recurring issues. By centralizing maintenance data, curators can prioritize tasks, comply with preservation standards, and coordinate seamlessly with exhibition planning, ensuring artifacts remain display-ready while safeguarding their historical value. This integration enhances operational efficiency and supports data-driven conservation strategies.

Sample Report 3

Maintenance Report Sunday, April 20, 2025

Artifact Type	Artifact Name	Condition of Artifact	Type of Maintenance	Date of Maintenance
Manuscript	Dead Sea Scroll	Good	Restoration	2018-09-14
	Dead Sea Scroll	Good	Cleaning	2017-10-21
Painting	Mona Lisa	Excellent	Repair	2007-05-30
	Mona Lisa	Excellent	Repair	2022-01-13
	Mona Lisa	Excellent	Cleaning	2006-04-12
Sculpture	Roman Coin Set	Good	Preservation	2012-07-11
	Egyptian Sarcophagus	Excellent	Cleaning	1996-05-23
	Terracotta Warrior	Need Repair	Inspection	2004-06-18

Sunday, April 20, 2025 Page 1 of 1

Figure 4.3.3.1: Maintenance Report

CHAPTER 5 CONCLUSION

5.1 System Weaknesses

The current system has several weaknesses, including limited scalability and difficulty in handling large volumes of data. Performance issues may arise as the database grows, especially when running complex queries or generating reports. Additionally, the system lacks sufficient user role management, leading to potential access control issues. Some features may also be unintuitive for less technical users, requiring additional training or support.

Another weakness is the absence of automated data backup and recovery processes, which increases the risk of data loss in case of system failures. Furthermore, the current interface could be improved for better user experience, as navigation between different forms and reports could be made more seamless. These limitations hinder the system's overall effectiveness in handling larger, more dynamic datasets.

5.2 Future Improvements

Future improvements should focus on enhancing system scalability by optimizing database structure and improving query performance. Implementing role-based access controls will help better manage user permissions and prevent unauthorized access to sensitive data. Additionally, introducing automated data backup and recovery mechanisms will increase data security and minimize downtime in case of system failure.

The user interface could also benefit from an overhaul to make it more intuitive and user-friendly, reducing the need for extensive training. A more dynamic and responsive design would enhance the overall experience. Finally, integrating new features such as data analytics or visualization tools could provide users with better insights and decision-making capabilities, improving the system's overall functionality and utility.

REFERENCE

[1] K. Rapp, "List of Museum Jobs Their Roles, Responsibilities, and More.," *ACME Ticketing*, Jul. 03, 2019. https://acmeticketing.com/list-of-museum-jobs/